

Unlocking new mining production through Advanced analytics

Business Challenge:

- One of the world's leading mining companies in North America
- Wanted to increase productions To maintain growth in its Americas copper operations
- Faced challenges as the existing portfolio of mature mines and aging technology would lead to significant cash outlays and lengthy permitting to open new mines.

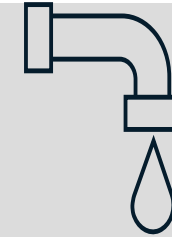
Solution:

- The solution involved enhancing operational efficiency through advanced analytics to boost copper production from existing mines and utilize technology to access untapped copper resources.
- Client adopted a consulting-driven assessment approach to optimize decision-making across the entire value chain.
- Modern mining is a complex and technical undertaking. Technologies such as cloud computing, centralized data warehouses, wireless mesh networks, and IoT sensors enabled Hexaware to bring data closer to operational sites and leverage historical data for insights.
- Hexaware utilized its expertise in data engineering and advanced analytics to unlock copper resources primed for production.

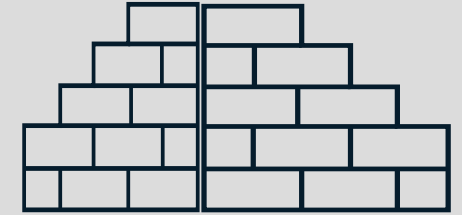
200M
Increase in pounds
In annual copper
productions across mines

\$350M+
(up to \$500M)
EBITDA improvements by
scaling analytics

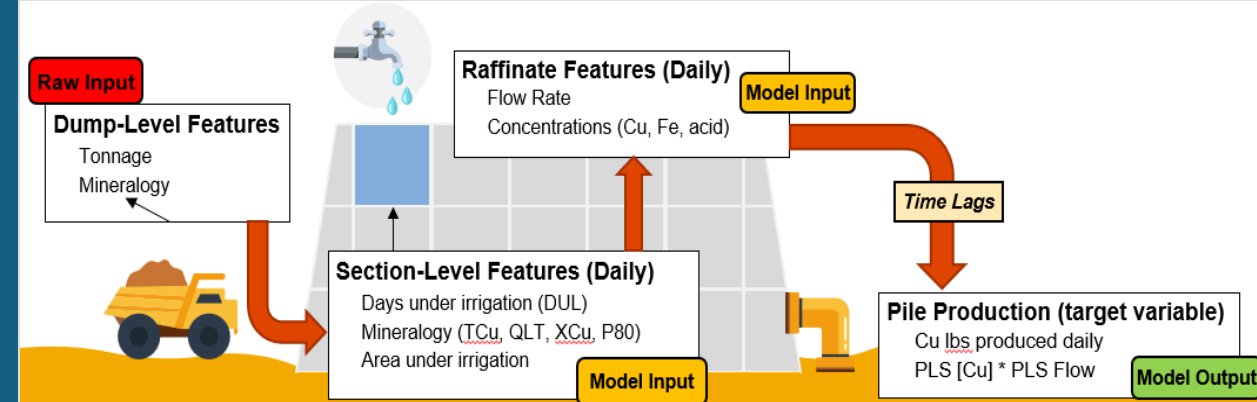
\$1.5-2B
The cost of a new
processing facility that
could be avoided



How much copper does a
leach pile produce



How much copper is left in
the leach pile



Technology Stack

- **DevOps** : Trunk-based development with continuous deployments.
- **DataOps** : loosely coupled automated data deployments
- **Document as Code**: Versioned documentation stored in repo that sources data discovery platforms like castor
- **Data Quality**: Data pipelines with automated data quality checks thought the pipeline
- **Modern tools** : Clout native, scalable tools such as Snowflake, Dbt, Dagster, Python, AzureML